MCGILL RETAIL GEN AI HACKATHON · FALL 2026

From Idea to Agent

Building on AWS

An intro to agentic AI and the AWS services you can build with

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AWS Virtual Workshop · Wed Sep 30, 2026 · 5:30 to 6:30 PM EDT

00 · AGENDA

What we'll cover

01

What an agent really is

Assistant versus agent, and the levels of autonomy

02

Amazon Bedrock

Your model foundation, live in the AWS console

03

From demo to production

AgentCore: the platform to run agents at scale

04

Build faster

Strands and Kiro, with short videos

05

The finals

Your AWS environment, mentors, and the jury

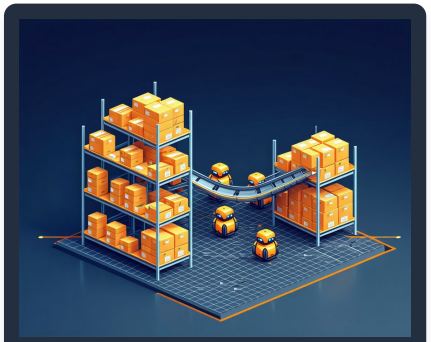
06

Start on your own

The new AWS Free Tier and free training

Machine learning is in Amazon's DNA

AWS did not arrive at AI last year. Amazon has built with machine learning for over two decades.



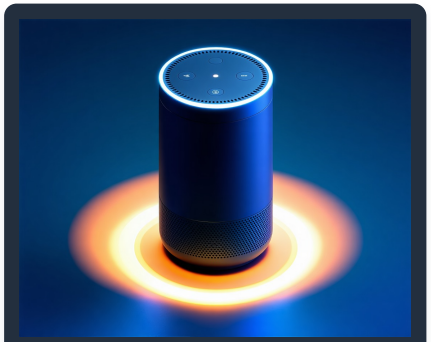
Fulfillment robotics

1M+ robots across our network



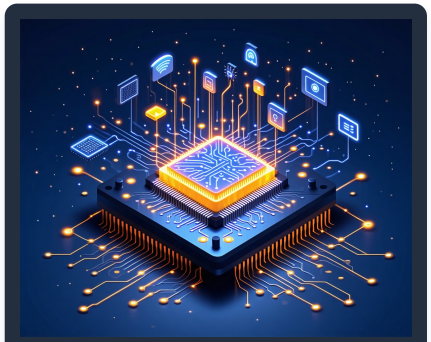
Prime Air

Autonomous drone delivery



Alexa

Billions of interactions a week



Generative AI

Amazon Bedrock and Nova

We turned that experience into services any developer can call in a few lines of code. **When you build this week, you stand on 20 years of innovation.**

Assistant vs. Agent

AI Assistant

- **Reactive** responds to your input
- **One turn at a time** you drive every step
- **Mostly retrieves and answers** limited to RAG
- **Needs explicit commands** think chatbot

AI Agent

- **Proactive** initiates actions toward a goal
- **Reasons and plans** breaks a goal into steps
- **Uses tools and data** calls APIs, observes results
- **Minimal hand-holding** it acts to get work done

An assistant answers questions. An agent gets work done.

Agents work at different levels of autonomy

L1

Analytical

Gathers info and gives insights. Changes nothing.

GREAT FOR
Research, trend analysis

L2

Logical

Makes calculated changes within set limits.

GREAT FOR
Pricing, inventory optimization

L3

Transactional

Takes independent digital actions.

GREAT FOR
Process an order, email a customer

L4

Physical

Controls the physical world.

GREAT FOR
Fulfillment, logistics robotics

Most agents you build over a weekend live at L2 to L3. That is plenty to show real autonomy.

Amazon Bedrock: your model foundation

Many top models, one API

Amazon Nova, Anthropic Claude, Llama, Mistral and more. Swap models without rewriting your app.

Your data stays yours

Private by default and never used to train the base models.

Knowledge Bases and Guardrails

Ground answers in your own data, and add safety controls.



LIVE DEMO

Open the AWS console

Browse the Bedrock model catalog

Run one prompt in the playground

aws.amazon.com/bedrock

From demo to production: Amazon Bedrock AgentCore



1. Model

reasons about the goal

+



2. Context

memory, knowledge, web search

+



3. Tools

call APIs, take action

+



4. Environment

run many agents safely at scale

AgentCore provides the environment: runtime, memory, gateway to tools, identity, and observability. Any framework, any model.

2.1x faster development

75% less infrastructure time

5.2x faster deployment

Jump straight to agents: AgentCore CLI quickstart · Source: Signal65, vs custom development

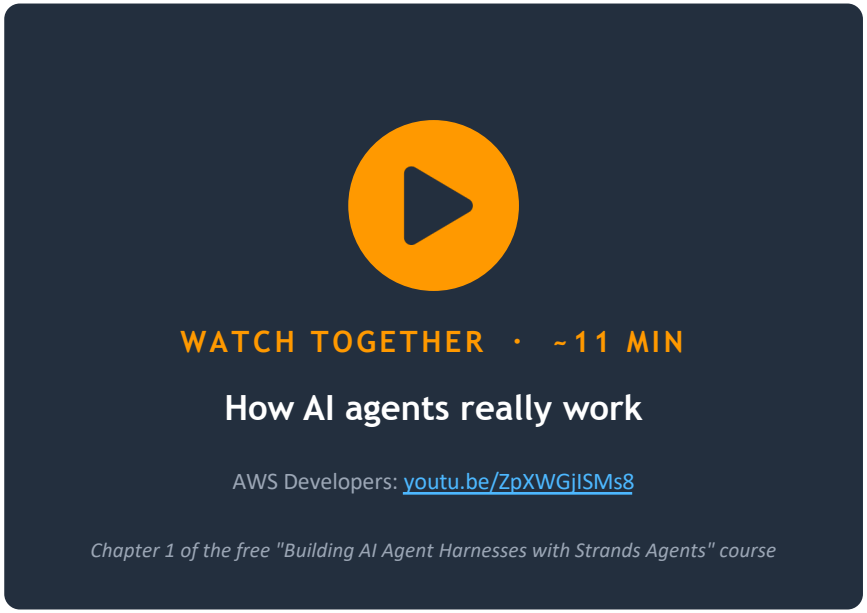
04 · BUILD FASTER

Build faster: Strands Agents SDK

- **Open-source SDK** Python or TypeScript
- **A working agent in a few lines** pick a model, add tools, run the loop
- **A fast on-ramp** ideal for a hackathon prototype

strandsagents.com

Keep learning on your own: the AWS Developers channel
youtube.com/@awsdevelopers

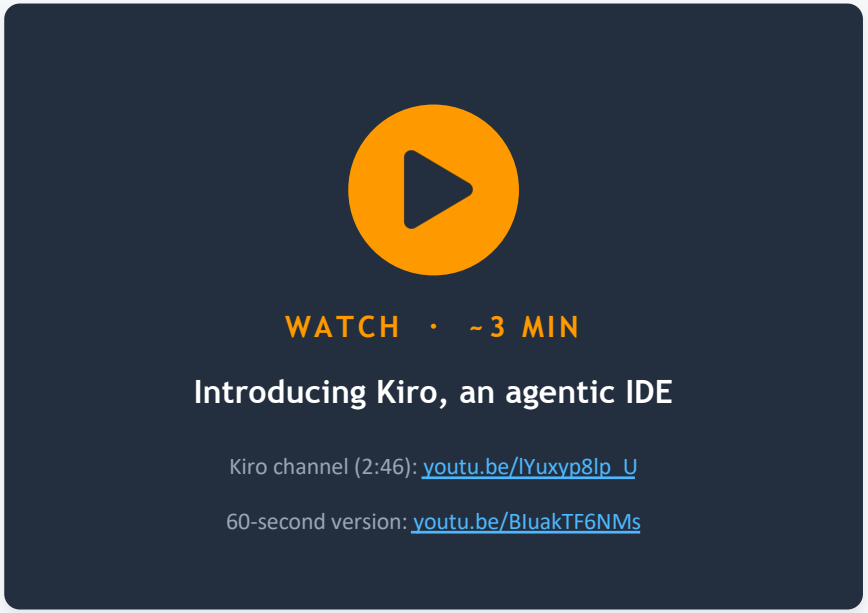


04 · KIRO

Kiro: your agentic coding partner

- **AWS agentic IDE** built on the editor you know
- **Spec-driven** turns a prompt into requirements, design, and tasks, then writes code
- **Great for mixed-skill teams** non-coders help shape the spec
- **Start on the free tier** 50 credits with a social or Builder ID login, paid plans from $20/mo

kiro.dev · Kiro + AgentCore how-to: AWS blog



05 · THE FINALS · OCT 15-16

Your AWS environment at the finals

- **One AWS account per finalist team** build your full solution on AWS
- **AWS services at no charge** for the two days of the event
- **Activated the morning of Oct 15** ready when hacking begins



The account is wiped

Every account is completely destroyed at the end of the event on October 16. Nothing survives.

Keep your code on GitHub
Keep data and notes on your laptops

Treat the AWS account as a powerful two-day workspace, not as storage. Plan for the wipe from the start and you will not lose a thing.

Mentors on site, and how the jury sees AWS

5 AWS Solutions Architects

- On site as mentors for the full two days
- Ask us anything: architecture, service choice, cost, security, debugging
- Live guidance and answers whenever you are stuck

We guide and unblock you. We do not code your solution for you.

AWS leaders on the jury

How well you use AWS is an important factor in your score.



Manon Yelle

SA Manager, Quebec (L7)

AWS WWPS Canada



Lisa Richardson

Enterprise Sr. Leader (L7)

AWS Canada

06 · START ON YOUR OWN

The new AWS Free Tier

AWS just reimagined getting started. It is simpler and safer, which is perfect for experimenting on your own.



Sign in fast

Use Google, GitHub, or Apple. For most new users, no credit card to start.



$100 in free credits

Build immediately in your first project. Earn extra credits as you deploy.



Wire up a coding agent

Paste the setup prompt and your agent deploys resources for you.



Spend limit keeps you safe

On a paid plan, set a limit from $20/mo. Hit it and the project pauses, no surprise bills.

Start: **aws.amazon.com/free** Read about the new experience: the AWS blog

06 · KEEP LEARNING

Keep learning, for free



AWS Student Rewards

Verified students: 12 months of Skill Builder Premium (900+ courses, hands-on labs), AWS credits, and a certification exam voucher.

builder.aws.com/learn/students



AWS Skill Builder

1,000+ free resources, including GenAI and agents learning plans, plus SimuLearn.

skillbuilder.aws



GenAI learning path

Curated courses on generative AI and Amazon Bedrock: Knowledge Bases, Agents, Guardrails.

aws.amazon.com/training/learn-about/generative-ai



AWS Certified AI Practitioner

An achievable first certification to put on your resume.

aws.amazon.com/certification

Go build. We're here to help.



Reach for agents, not just chatbots

Show reasoning and action, the L2 to L3 sweet spot.



Lean into AWS services

Bedrock, AgentCore, Strands, Kiro. Use them and explain them.



Use your mentors

Five AWS SAs on site for two days. Ask us anything.

You have the tools, the mentors, and a free path to keep going.

I cannot wait to see what you build.

Vincent Rioux
Senior Solutions Architect, AWS Canada

Questions welcome in English or French